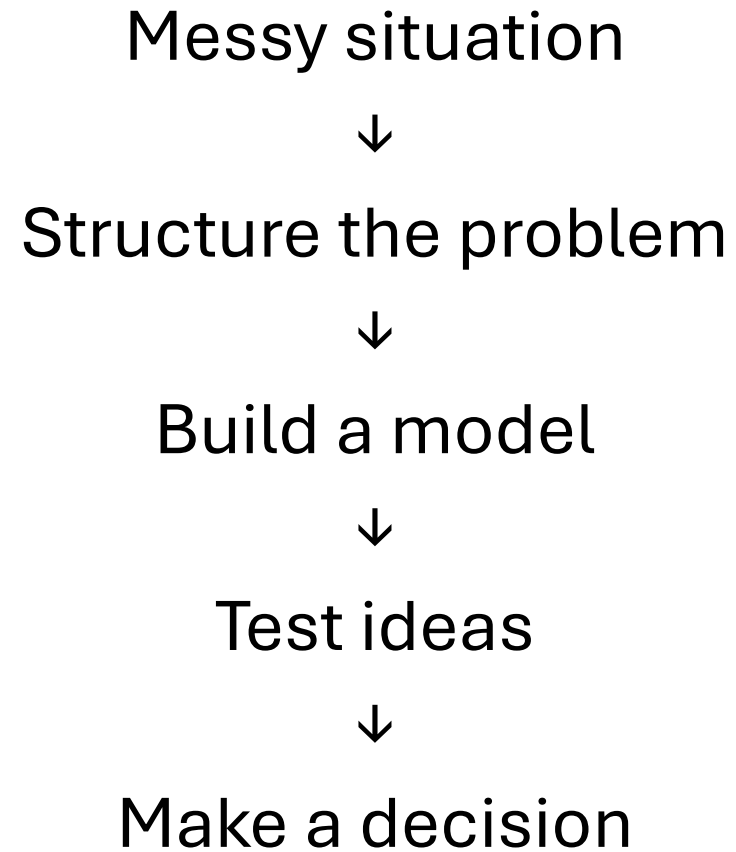


Problem Structuring and Modelling

Emergency Department

messy problem → model → decision

Session Plan



Rules

- 1) Think first
- 2) Predict before seeing results
- 3) No “right” model at the start

Scenario

A hospital is reviewing staffing levels in its Emergency Department (ED).

Recently:

- Waiting times have increased
- Staff feel under pressure
- Managers are concerned about costs

Patients arrive throughout the day needing treatment. Some cases are quick to resolve, while others require more time and attention.

The department is staffed by a team of doctors and nurses. Staff work together to assess and treat patients.

Hospital management want to understand:

- What changes might improve performance while controlling cost.

They are open to analysis but are not yet sure what kind of modelling or data is needed.

You have been asked to investigate.

Part A

Problem Structuring

Part A

Worksheet Part A

In groups discuss:

- What decisions might be made?
- What would we need to know?
- Sketch the system

Time: 20 minutes

Share back

- Share what we have found from Part A

Time: 10 minutes

Part B

Prediction

Prediction Phase

Worksheet Part B

In groups discuss:

- What happens if staffing increases?
- What might still cause delays?
- What kinds of changes might improve the system?

Time: 10 minutes

Conceptual Thinking

Arrivals → ? → Treatment → Exit

What might happen between arrival and treatment?

Part C

Building the Model

First Model

Add Triage

- Gather more information about a patient?

Variation

Some cases take longer than others

Add variability.

Patient Types!

- critical
- urgent
- standard

Fairness

Should everyone be treated strictly in arrival order?

Add in priority queue

Extreme Case

What if something very serious arrives during treatment?

Add in pre-emption

Should anyone skip triage?

Major thing that we haven't modelled

Cost

We would require cost to do any serious analysis!

Part D

Results

What is the optimal number of doctors?

Define optimal:

- minimise critical breaches
- keep urgent breaches low
- avoid excessive cost
- maintain reasonable utilisation

You already have the data from the simulation. For each doctor level:

- critical breaches
- urgent breaches
- standard breaches
- utilisation

So, the decision rule becomes something like:

Choose the smallest number of doctors that keeps critical breaches ≈ 0 and urgent breaches acceptable.

Results from Simulation

You can access the full simulation that we have just built on the module.

Use this to fill out Part D and work out how many doctors are acceptable.

Also consider what costs would be needed for this simulation to be more realistic!